



ASIAN
INSTITUTE OF
MANAGEMENT

Machine Learning 3

Transformer Architecture

Session 15 and 16

MSDS 2026

25 May 2026

ASIAN
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MANAGEMENT

Class Administrative Matters

Course Schedule

Session 15 and 16 (May 25) – Transformer Architecture

Session 17 and 18 (May 29) – Journal Scan Presentation

Session 19 and 20 (June 2) – Final Exam

Final Project Presentation (June 10)



Class Administrative Matters

Course Deliverables

E8 – Transformer Architecture (June 3)

MP2 – Deep Learning using RNNs and Transformer (June 8)

Final Project Presentation Poster (June 9)



Session 15 and 16

Session 15 and 16 – Transformer Architecture

Gameplan

Session 15

NLP using RNNs

Transformer Fundamentals

Session 16

Machine Translation using Transformer Architecture

NLP using RNNs

NLP using RNNs

IMDB Sentiment Analysis Model Performances

For the *IMDB Movie Reviews* dataset, the model performances of the different iterations of machine learning models is:

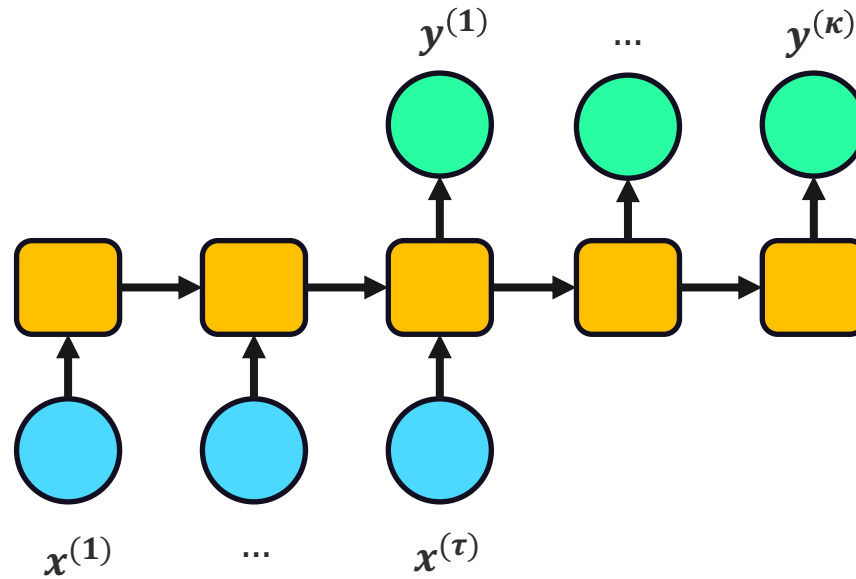
Model	Accuracy
Bag-of-Words	88.8%
Bi-Gram	90.2%
LSTM	83.2%
LSTM with Embedding	86.2%
LSTM with Pretrained Embedding	88.4%



NLP using RNNs

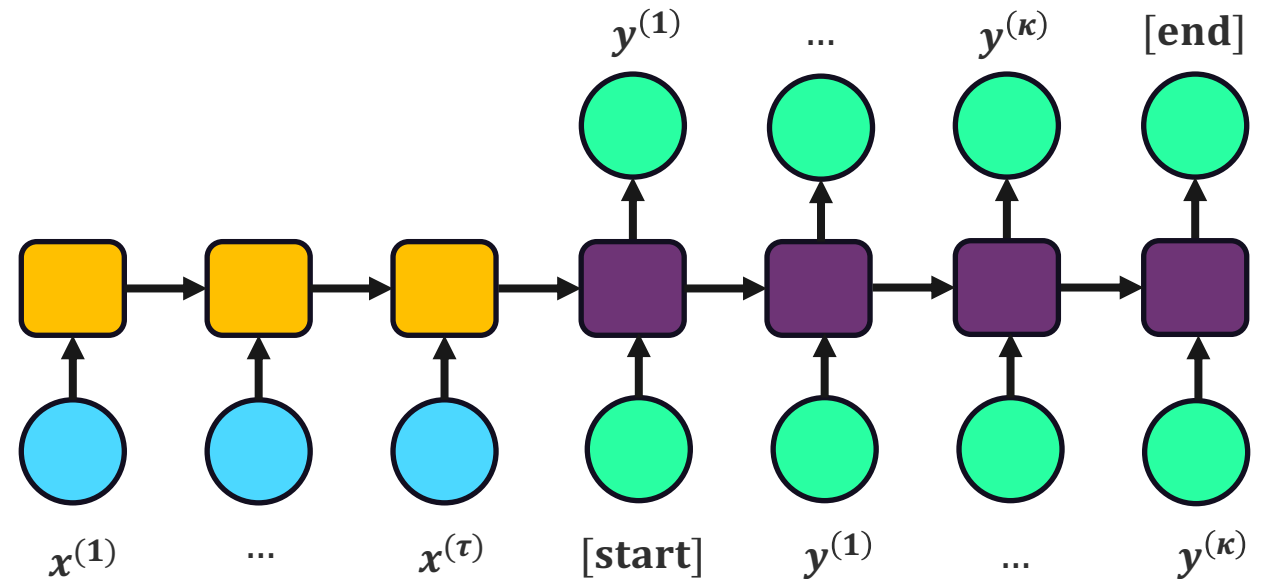
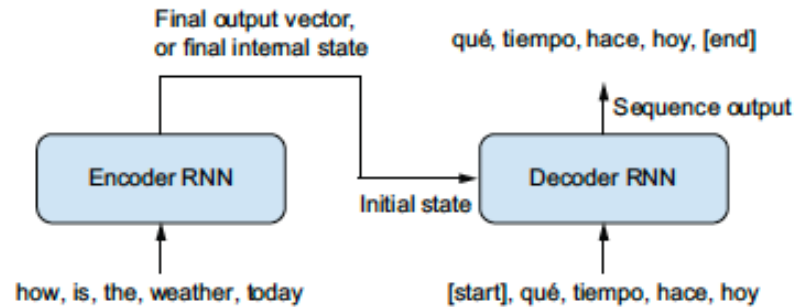
Sequential Modeling

Many-to-Many



NLP using RNNs

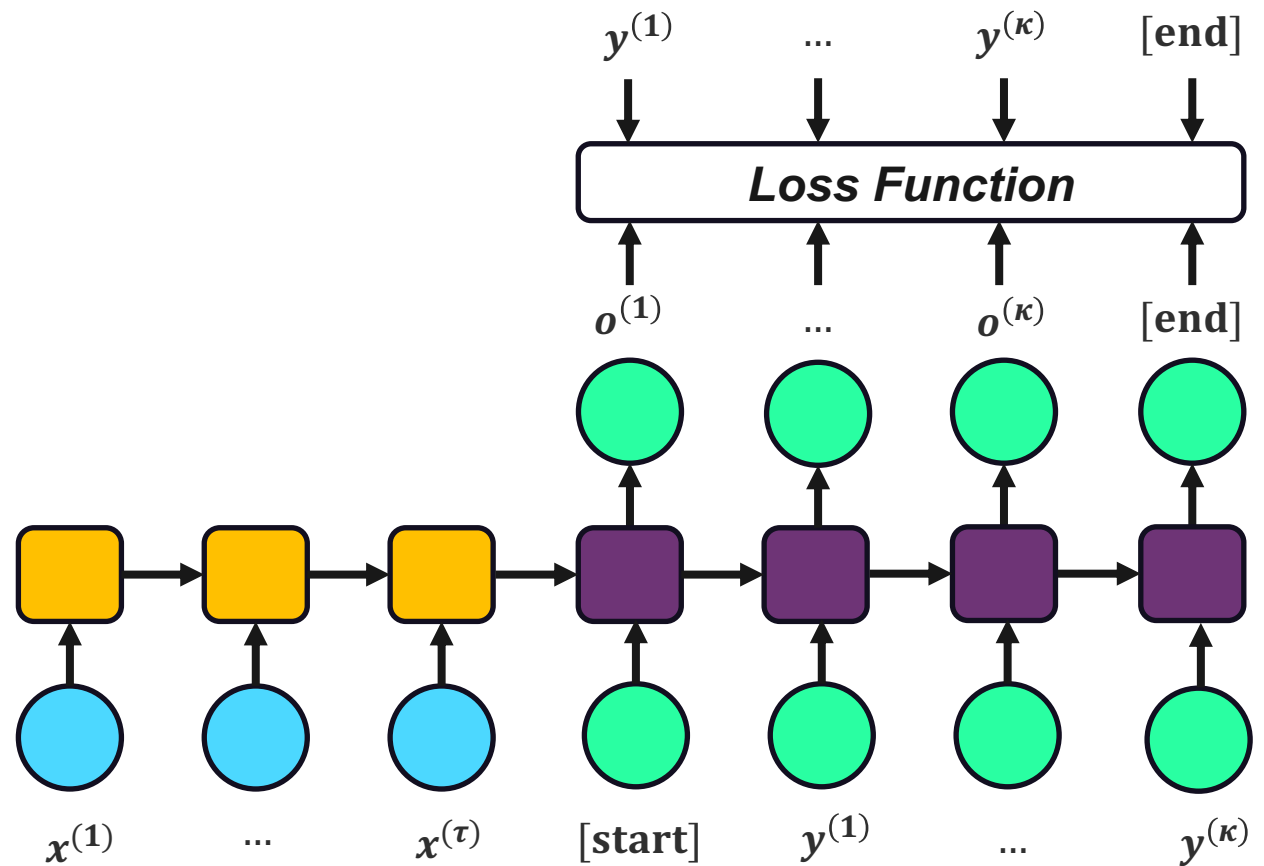
Sequence to Sequence RNN: Encoder-Decoder Architecture



NLP using RNNs

Sequence to Sequence RNN: Encoder-Decoder Architecture

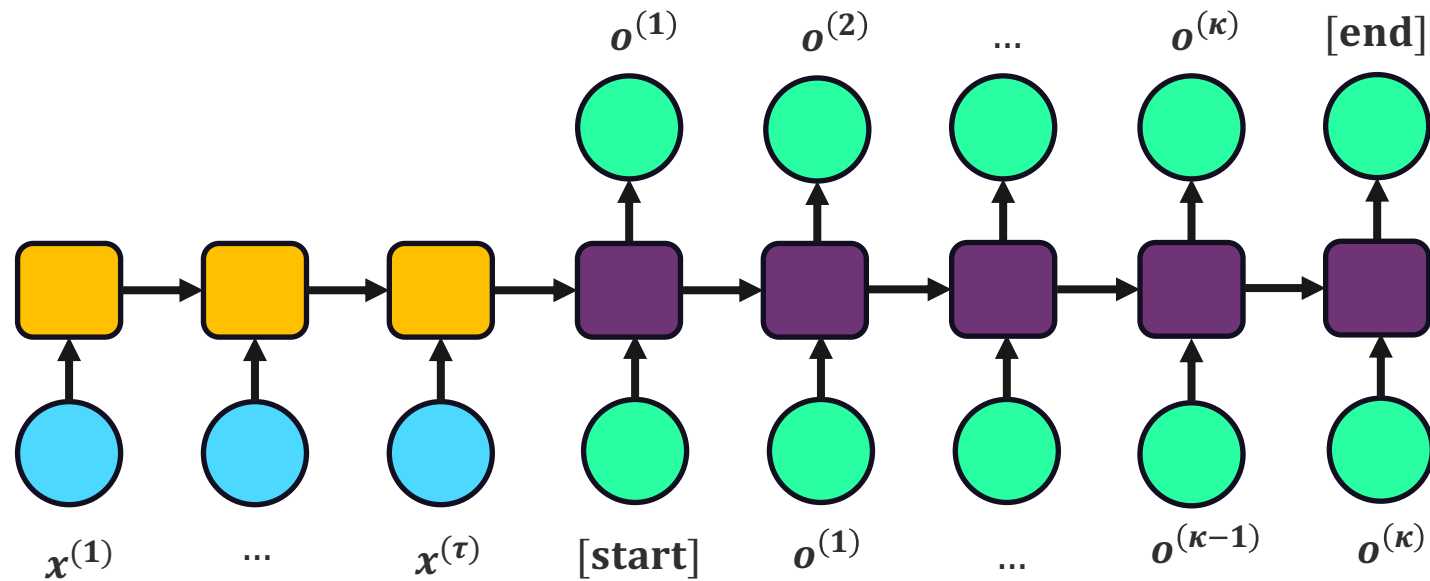
Model Training



NLP using RNNs

Sequence to Sequence RNN: Encoder-Decoder Architecture

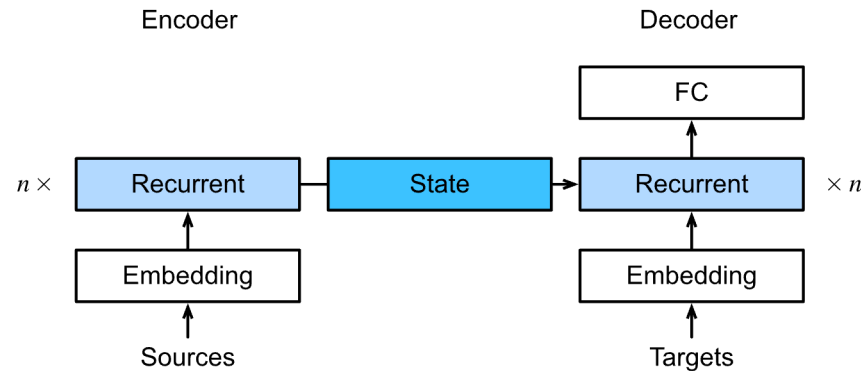
Model Deployment



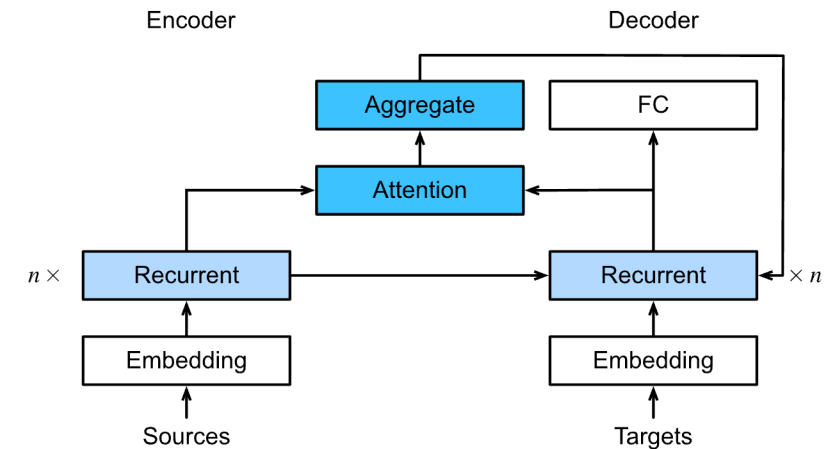
NLP using RNNs

The Bahdanau Attention Mechanism

RNN Encoder-Decoder



RNN Encoder-Decoder (with Attention)

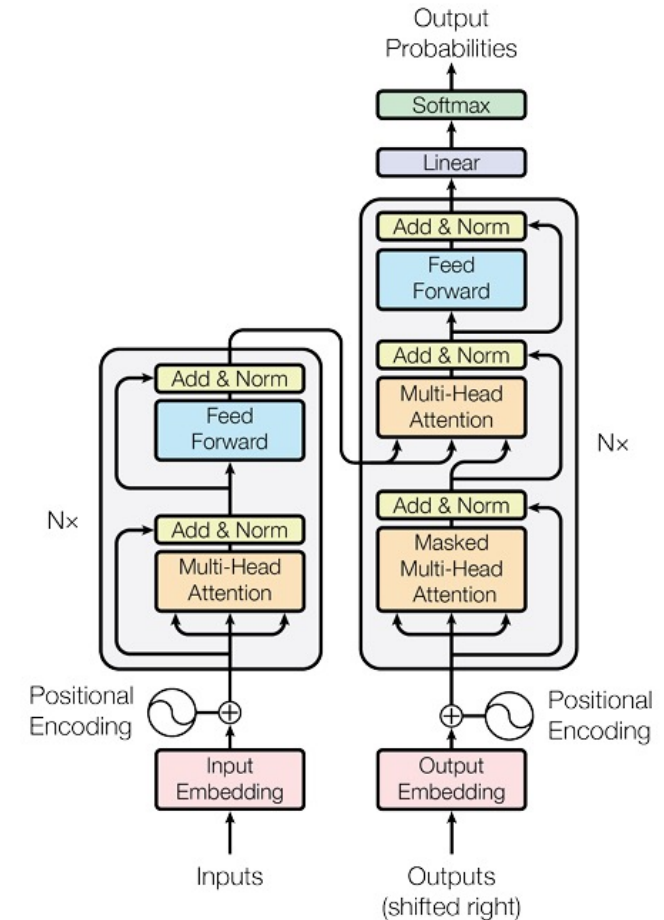


$$\mathbf{c}^{(t')} = \sum_{t=1}^{\tau} \alpha(\mathbf{s}^{(t'-1)}, \mathbf{h}^{(t)}) \mathbf{h}^{(t)}$$

NLP using RNNs

Attention is all you need

“In this work we propose the **Transformer**, a model architecture **eschewing** recurrence and instead relying entirely on **attention** mechanism to draw global dependencies between input and output.”



The Transformer – model architecture

Transformer Architecture



Transformer Architecture

1 Attention

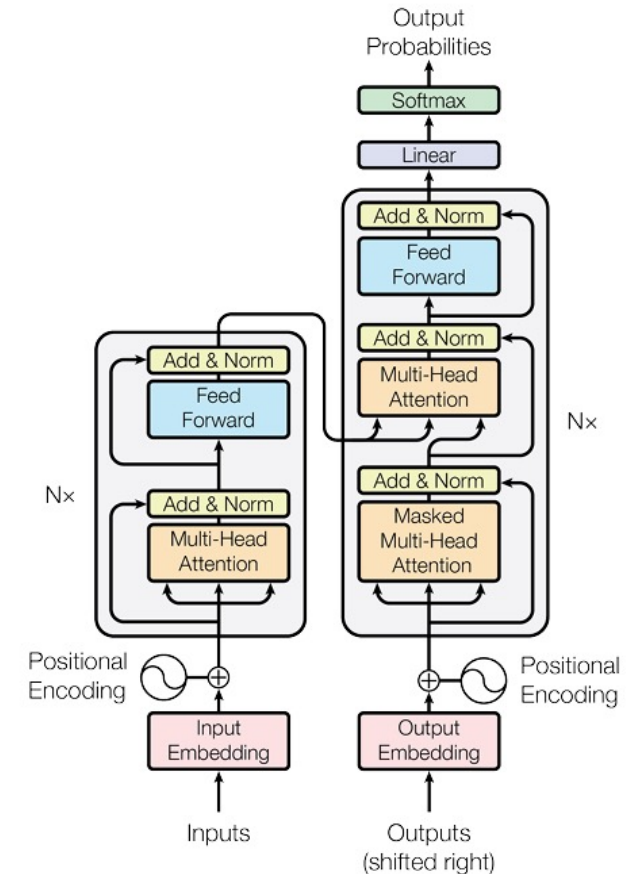
- Query, Key, Values – QKV
- Self-attention

2 Architectural Patterns

- Dense Projections
- Multi-head Attention
- Layer Normalization
- Residual Connections

3 Other Components

- Positional Encoding
- Causal Padding

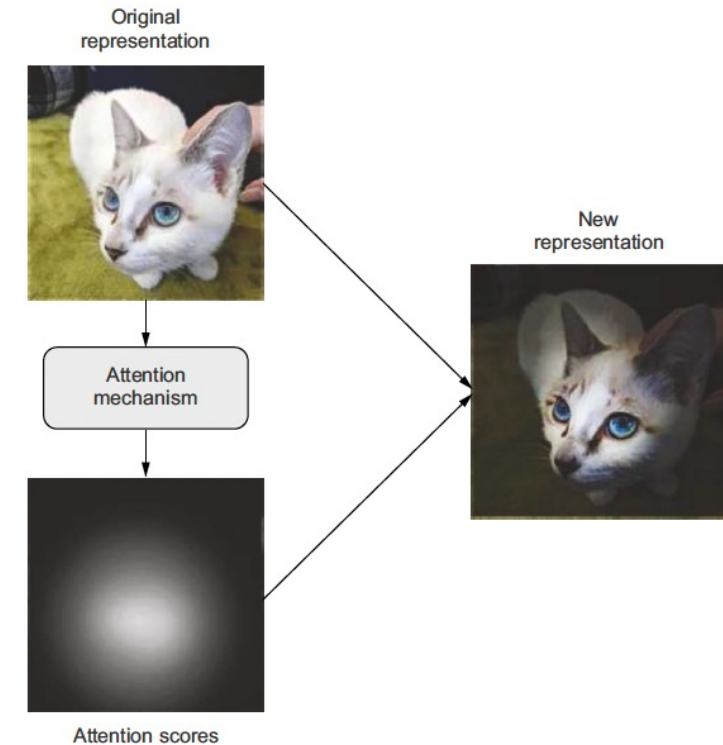


Transformer Architecture

Transformer Architecture

Attention Mechanism

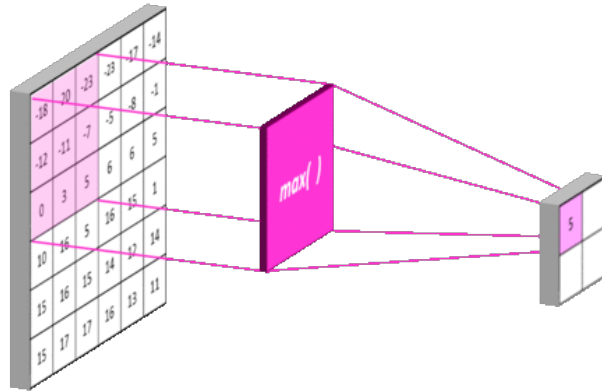
Key idea: “not all input information seen by a model is *equally important* to the task at hand, so models should **‘pay more attention’** to some features and *less* to other features”



Transformer Architecture

Attention-like Architectural Patterns

Max Pooling



TF-IDF

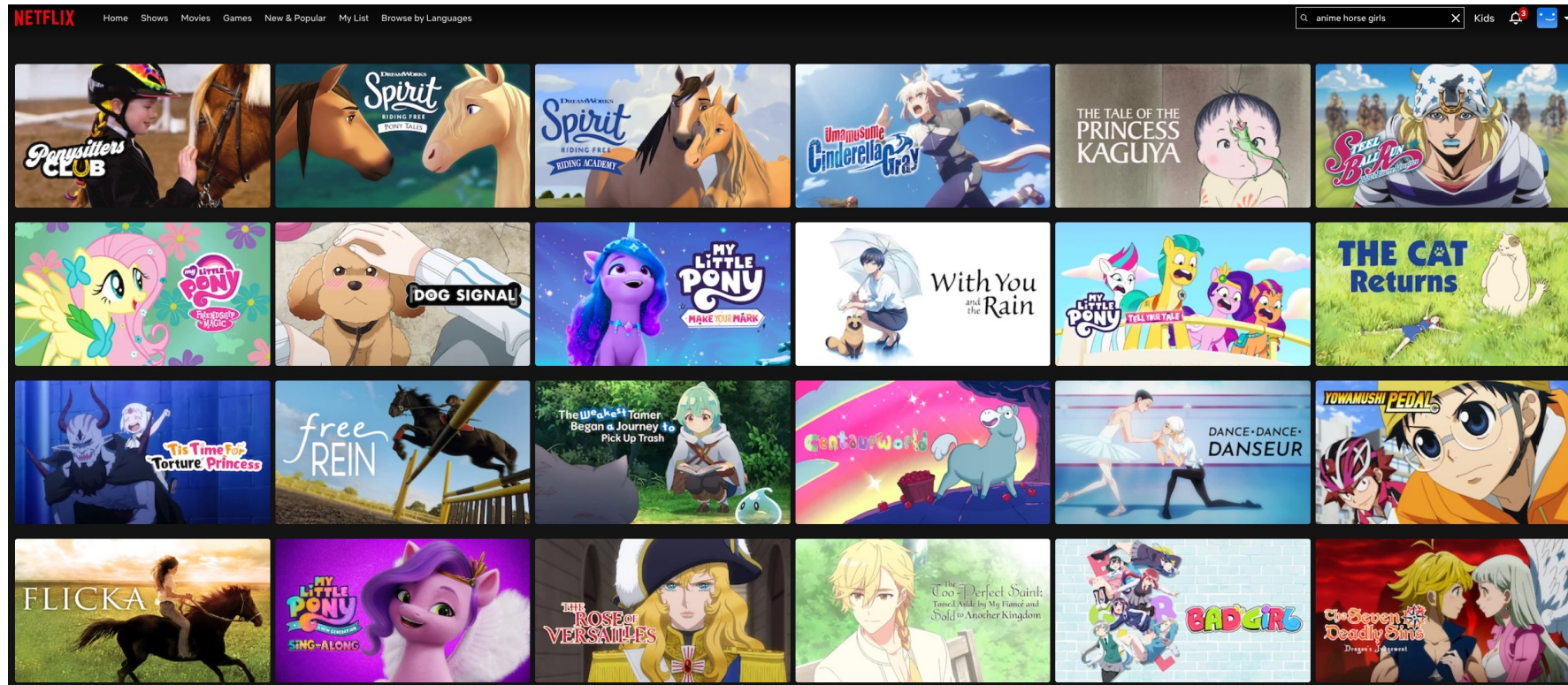
Term Frequency $\times$ Inverse Document Frequency

the movie is very fun and exciting

TF-IDF 0.00 0.00 0.00 0.68 0.25 0.00 0.68

Transformer Architecture

Queries, Keys, Values (QKV)



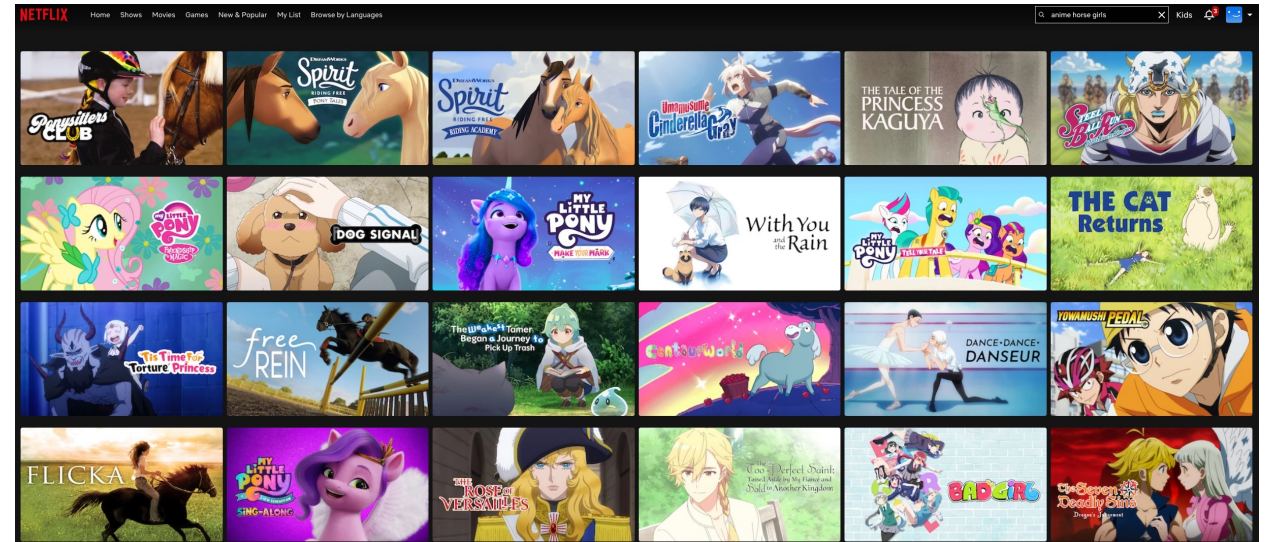
Transformer Architecture

Queries, Keys, Values (QKV)

Query: search terms

Keys: terms used to match query

Values: items to be retrieved



Transformer Architecture

Queries, Keys, Values (QKV)

Query: search terms

Keys: terms used to match query

Values: items to be retrieved

Denote a database $\mathcal{D}$ which comprises of key-value pair items such that $\mathcal{D} = \{(\mathbf{k}_1, \mathbf{v}_1), \dots, (\mathbf{k}_m, \mathbf{v}_m)\}$, we define the attention over $\mathcal{D}$ of a query $\mathbf{q}$ as

$$\text{Attention}(\mathbf{q}, \mathcal{D}) = \sum_{i=1}^m \alpha(\mathbf{q}, \mathbf{k}_i) \mathbf{v}_i$$



Transformer Architecture

Queries, Keys, Values (QKV)

$$\text{Attention}(\mathbf{q}, \mathcal{D}) = \sum_{i=1}^m \alpha(\mathbf{q}, \mathbf{k}_i) \mathbf{v}_i$$

Query: target language

Keys: source language

Values: source language



	"I"	"will"	"bring"	"the"	"bag"	"to"	"you"
"Te"							
"traeré"							
"la"							
"bolsa"							
[end]							

Transformer Architecture

Queries, Keys, Values (QKV)

$$\text{Attention}(\mathbf{q}, \mathcal{D}) = \sum_{i=1}^m \alpha(\mathbf{q}, \mathbf{k}_i) \mathbf{v}_i$$

A common strategy is to ensure that the attention weights is normalized to 1:

$$\alpha(\mathbf{q}, \mathbf{k}_i) = \frac{\alpha(\mathbf{q}, \mathbf{k}_i)}{\sum_j \alpha(\mathbf{q}, \mathbf{k}_j)}$$



Transformer Architecture

Queries, Keys, Values (QKV)

$$\text{Attention}(\mathbf{q}, \mathcal{D}) = \sum_{i=1}^m \alpha(\mathbf{q}, \mathbf{k}_i) \mathbf{v}_i$$

Additionally, we ensure that the weights are non-negative, thus given any scoring function a

$$\alpha(\mathbf{q}, \mathbf{k}_i) = \frac{\exp(a(\mathbf{q}, \mathbf{k}_i))}{\sum_j \exp(a(\mathbf{q}, \mathbf{k}_j))}$$



Transformer Architecture

Transformer-style Attention: QKV

$$\text{Attention}(\mathbf{q}, \mathbf{k}, \mathbf{v}) = \text{softmax}\left(\frac{\mathbf{q}\mathbf{k}^T}{\sqrt{d_k}}\right) \mathbf{v}$$



Vaswani, Ashish, et al. "Attention is all you need." Advances in neural information processing systems 30 (2017).

Transformer Architecture

Self-Attention

Input Sequence: The train left the station on time

	the	train	left	the	station	on	time
the	1.0	0.3	0.1	0.5	0.2	0.1	0.1
train	0.3	1.0	0.6	0.3	0.8	0.1	0.2
left	0.1	0.6	1.0	0.1	0.6	0.1	0.1
the	0.5	0.3	0.1	1.0	0.3	0.1	0.2
station	0.2	0.8	0.6	0.3	1.0	0.2	0.2
on	0.1	0.1	0.1	0.1	0.2	1.0	0.5
time	0.1	0.2	0.1	0.2	0.2	0.5	1.0

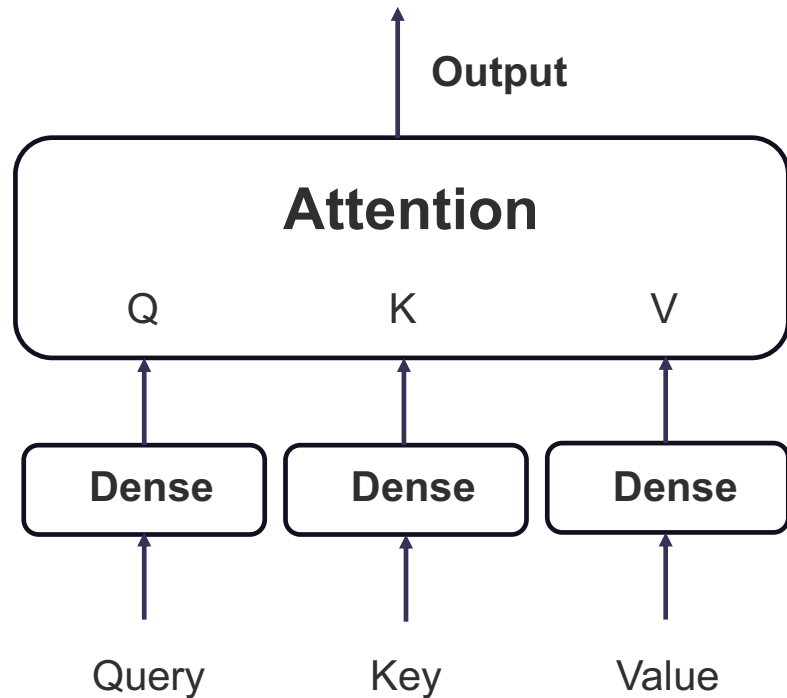
Attention scores

Self-attention produces ***context-aware*** token representations by attending the representation of a *token* using the representations of *related tokens* in the same sequence.



Transformer Architecture

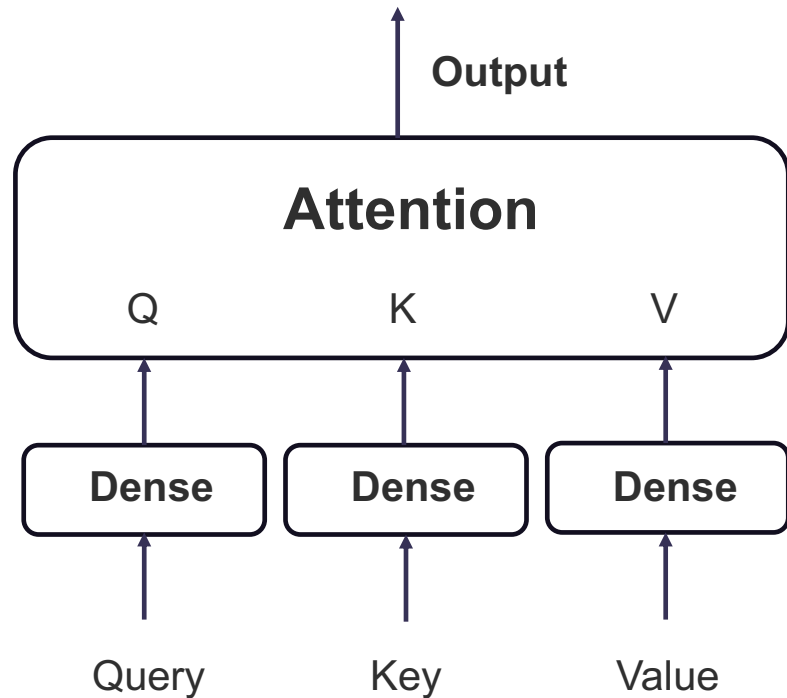
Dense Projections



Before computing attention scores, the query, key, and values are sent into separate dense neural networks, creating ***dense projections*** for the input tokens.

Transformer Architecture

Dense Projections

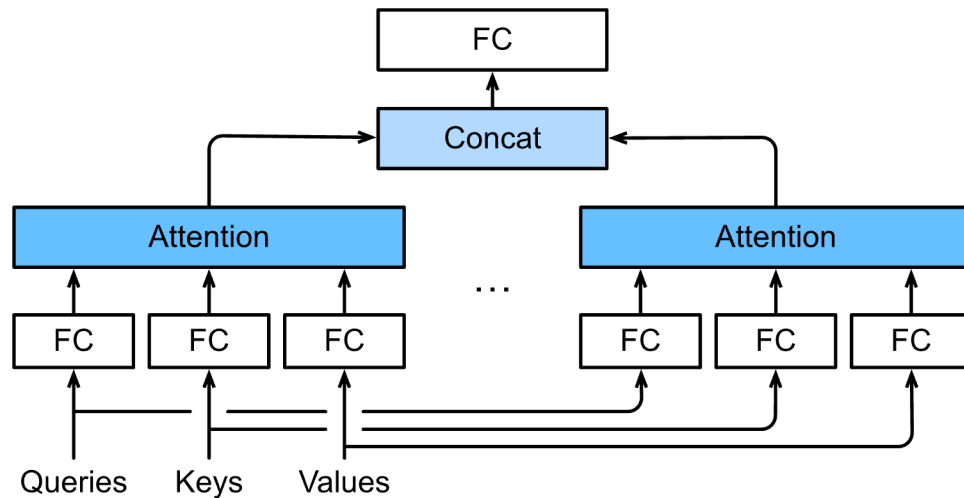


$$\text{Attention}(\mathbf{q}, \mathcal{D}) = \sum_{i=1}^m \alpha(\mathbf{q}, \mathbf{k}_i) \mathbf{v}_i$$



Transformer Architecture

Multi-Head Attention



Multi-headed attention allows each target word to attend to different parts of the source sequence in multiple ways.

Transformer Architecture

1 Attention

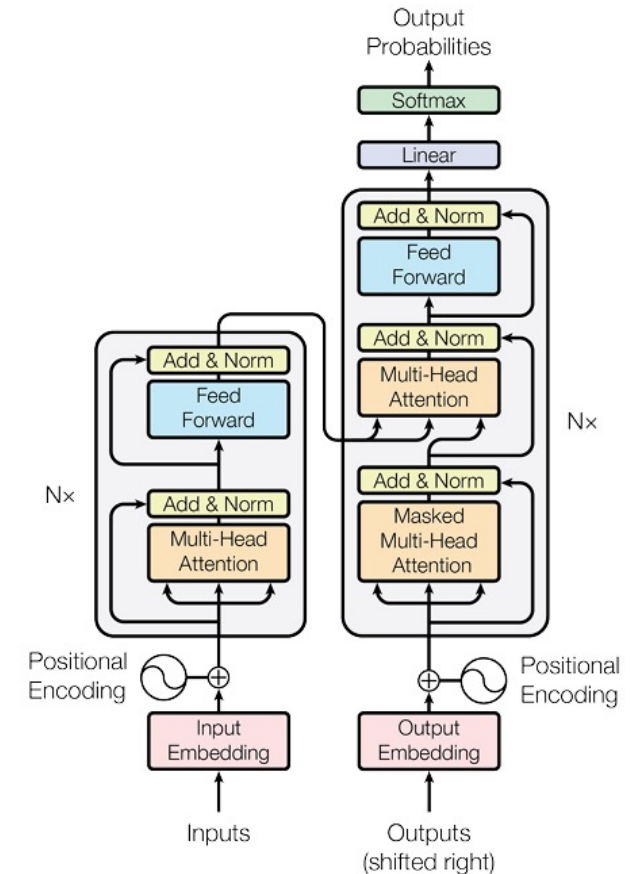
- Query, Key, Values – QKV
- Self-attention

2 Architectural Patterns

- Dense Projections
- Multi-head Attention
- Layer Normalization
- Residual Connections

3 Other Components

- Positional Encoding
- Causal Padding



Transformer Architecture

Transformer Architecture

Positional Encoding

Positional encoding incorporates information of the order in which the tokens appear in the sequence.



Zhang, Aston, et al. Dive into deep learning. Cambridge University Press, 2023.

Transformer Architecture

Positional Encoding

Sinusoidal Position Embedding

Let the input representation $\mathbf{x} \in \mathbb{R}^{n \times d}$ contain the d -dimensional embedding for n tokens.

The positional encoding $\mathbf{x} + \mathbf{p}$ is defined using a positional embedding matrix $\mathbf{p} \in \mathbb{R}^{n \times d}$, with $\mathbf{p}$ defined as:

$$p_{i,2j} = \sin\left(\frac{i}{10000^{2j/d}}\right)$$

$$p_{i,2j+1} = \cos\left(\frac{i}{10000^{2j/d}}\right)$$



Transformer Architecture

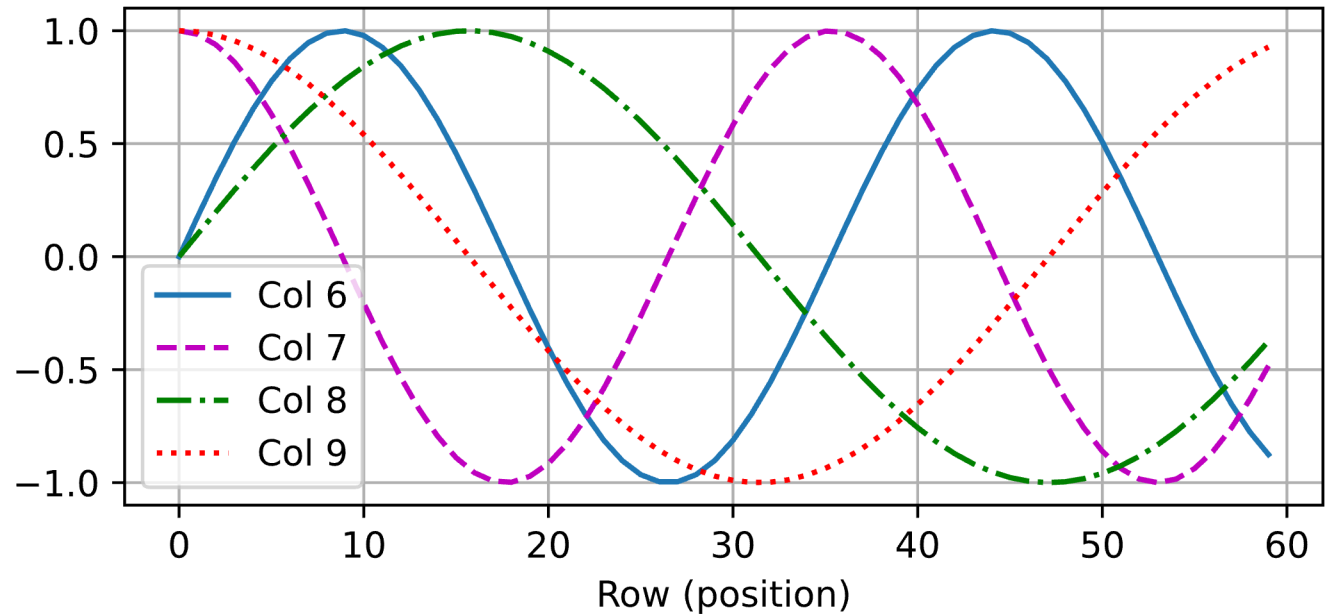
Positional Encoding

Sinusoidal Position Embedding

$$\mathbf{p} \in \mathbb{R}^{n \times d}$$

$$p_{i,2j} = \sin\left(\frac{i}{10000^{2j/d}}\right)$$

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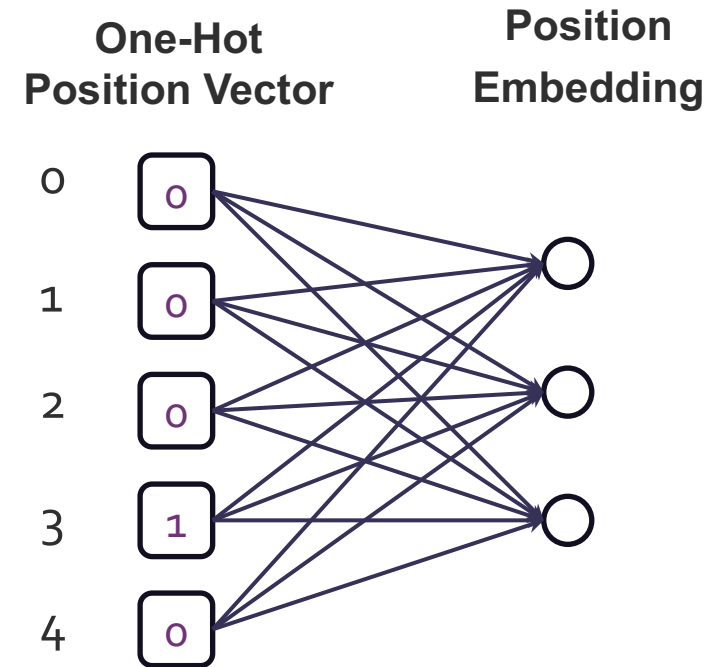


Transformer Architecture

Positional Encoding

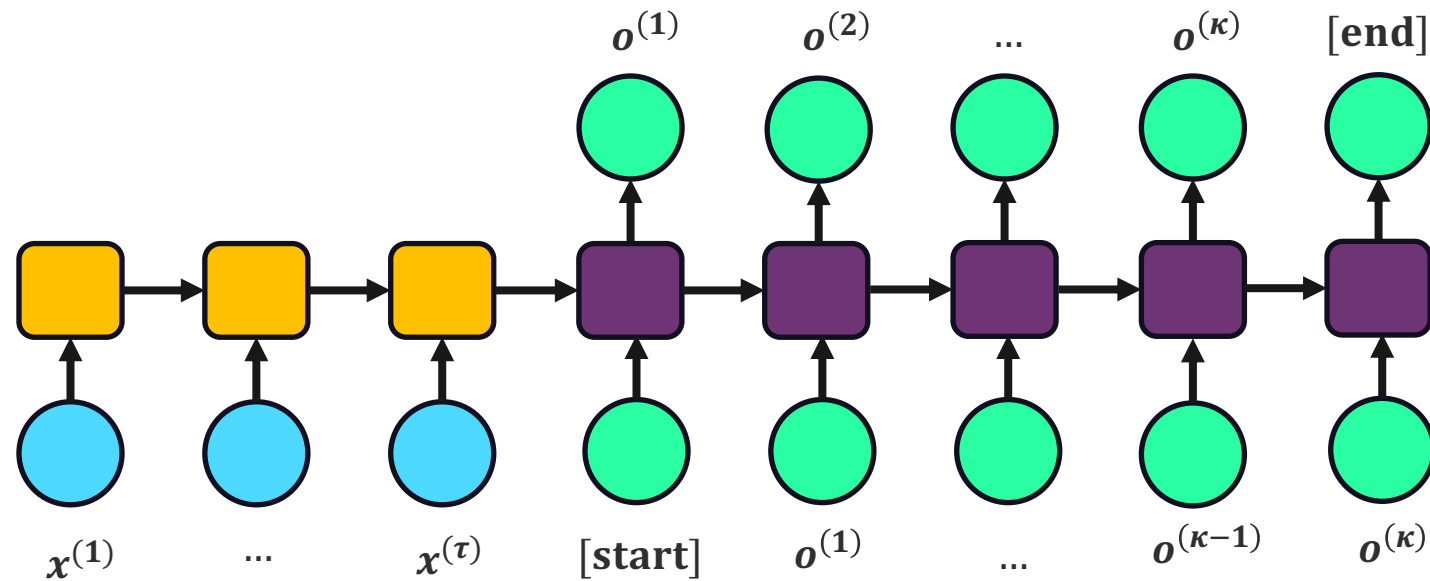
Learnable Position Embedding

i love ML !



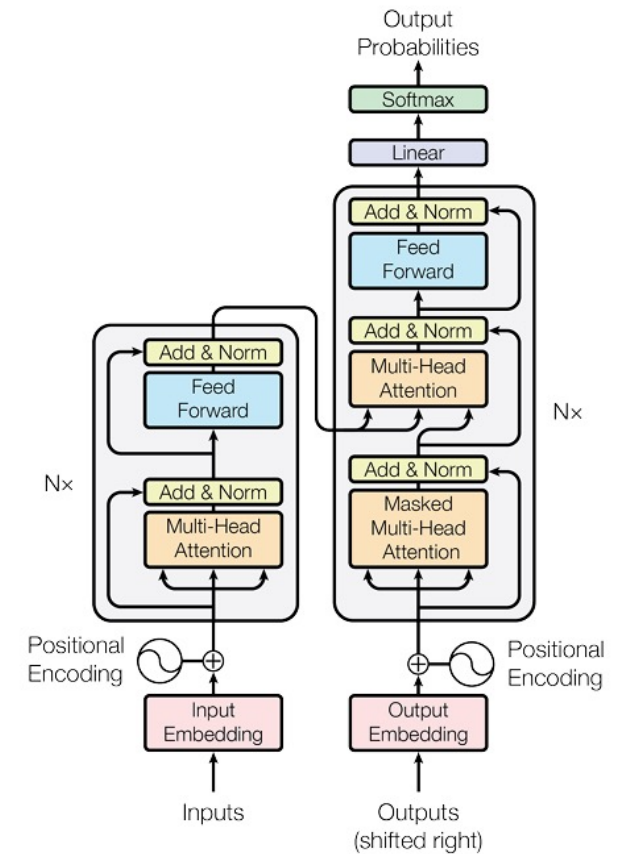
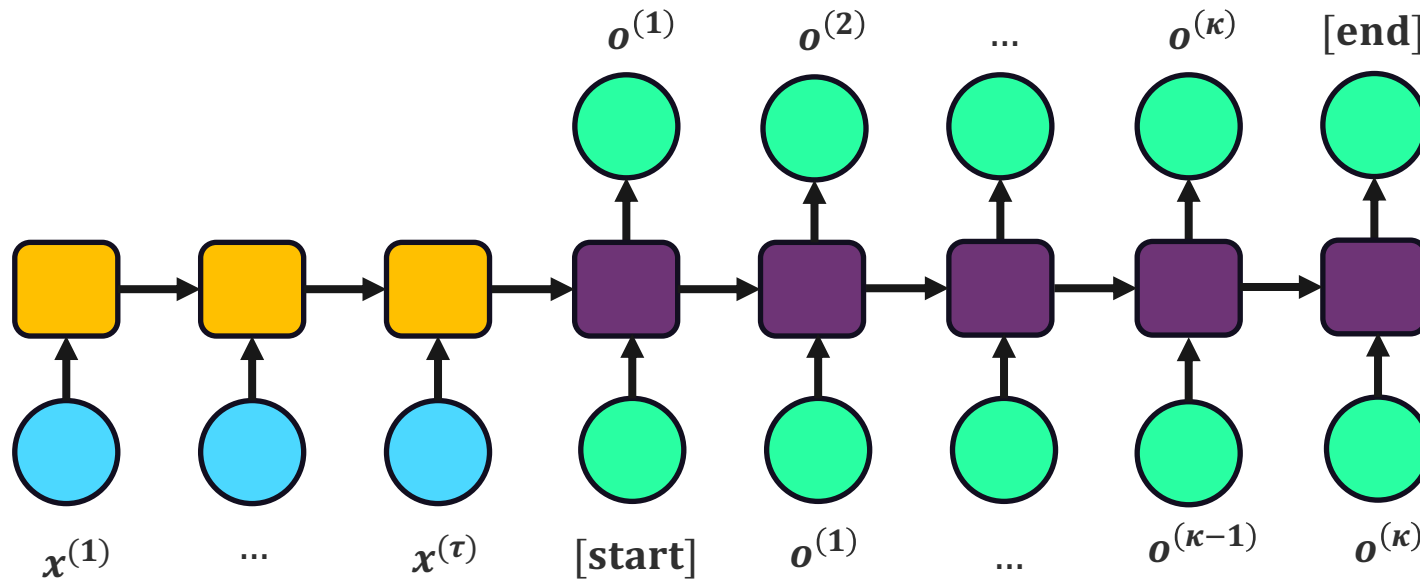
Transformer Architecture

Causal Mask



Transformer Architecture

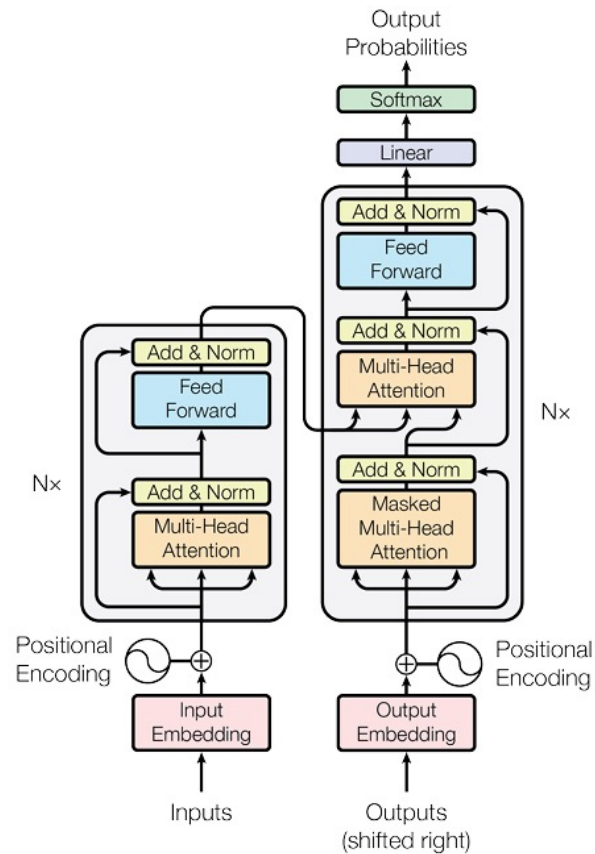
Causal Mask



Transformer Architecture

Transformer Architecture

Causal Mask



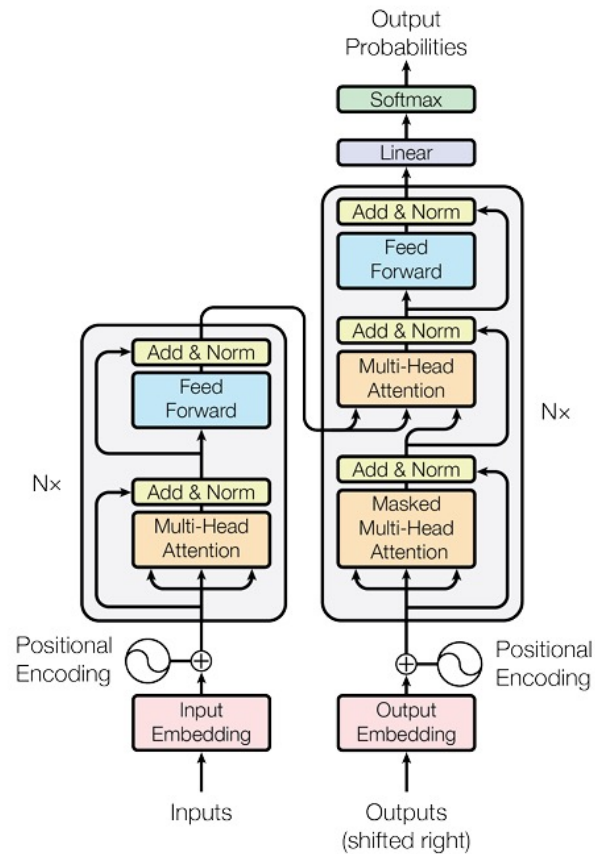
Problem: Transformers is inherently bidirectional

	the	train	left	the	station	on	time
the	1.0	0.3	0.1	0.5	0.2	0.1	0.1
train	0.3	1.0	0.6	0.3	0.8	0.1	0.2
left	0.1	0.6	1.0	0.1	0.6	0.1	0.1
the	0.5	0.3	0.1	1.0	0.3	0.1	0.2
station	0.2	0.8	0.6	0.3	1.0	0.2	0.2
on	0.1	0.1	0.1	0.1	0.2	1.0	0.5
time	0.1	0.2	0.1	0.2	0.2	0.5	1.0

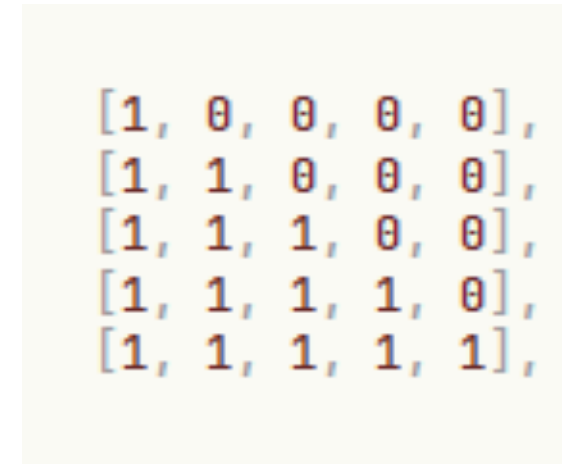
Attention scores

Transformer Architecture

Causal Mask



Solution: Use a **causal mask** to achieve one-directional information flow



Transformer Architecture

1 Attention

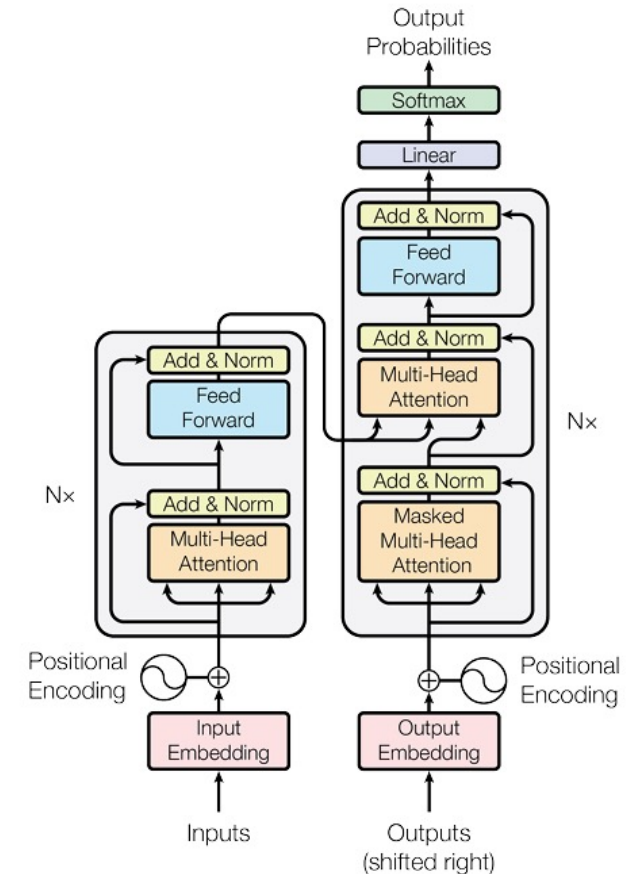
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